

## Chapter 5: The Internet and Its Uses

### The internet and the World Wide Web(WWW)

#### The differences between the internet and the World Wide Web (WWW)

- The word **internet** comes from **INTER**connected **NET**work.
- It is basically a worldwide collection of interconnected networks.
- The internet is actually a concept rather than something tangible (that is, something we can touch).
- It relies on a physical infrastructure that allows networks and individual devices to connect to other networks and devices.
- In contrast, the **World Wide Web (WWW)** is only a part of the internet that users can access using web browser software.
- The World Wide Web consists of a massive collection of web pages, and is based on the hypertext transfer protocol.
- World Wide Web is a way of accessing information using the internet.

| Internet                                                                                                                  | World Wide Web (WWW)                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• users can send and receive emails</li> </ul>                                     | <ul style="list-style-type: none"> <li>• it is a collection of multimedia web pages and other information on websites</li> </ul>   |
| <ul style="list-style-type: none"> <li>• allows online chatting (via text, audio and video)</li> </ul>                    | <ul style="list-style-type: none"> <li>• http(s) protocols are written using hypertext mark-up language (HTML)</li> </ul>          |
| <ul style="list-style-type: none"> <li>• makes use of transmission protocols (TCP) and internet protocols (IP)</li> </ul> | <ul style="list-style-type: none"> <li>• uniform resource locators (URLs) are used to specify the location of web pages</li> </ul> |
| <ul style="list-style-type: none"> <li>• it is a worldwide collection of interconnected networks and devices</li> </ul>   | <ul style="list-style-type: none"> <li>• web resources are accessed by web browsers</li> </ul>                                     |
|                                                                                                                           | <ul style="list-style-type: none"> <li>• uses the internet to access information from web servers</li> </ul>                       |

### Uniform resource locators (URLs)

**Uniform resource locators (URLs)** are text addresses used to access websites.

A URL is typed into a browser address bar using the following format:

**protocol://website address/path/file name**

- The **protocol** is usually either **http** or **https**.
- The **website address** is:
  - » **domain host** (www),
  - » **domain name** (website name),
  - » **domain type** (.com, .org, .net, .gov, for example),

» and sometimes **country code** (.uk, .de, .cy, for example).

- The **path** is the web page, but is often omitted and it then becomes the root directory of the website (see example below).
- The **file name** is the item on the web page. For example:  
<https://www.hoddereducation.co.uk/ict>

## HTTP and HTTPS

- **Hypertext transfer protocol (http)** is a set of rules that must be obeyed when transferring files across the internet.
- When some form of security (for example, SSL or TLS) is used, then this changes to **https** (you will often see the green padlock in the status bar as well).
- The 's' stands for secure, and indicates a more secure way of sending and receiving data across a network (for example, the internet).

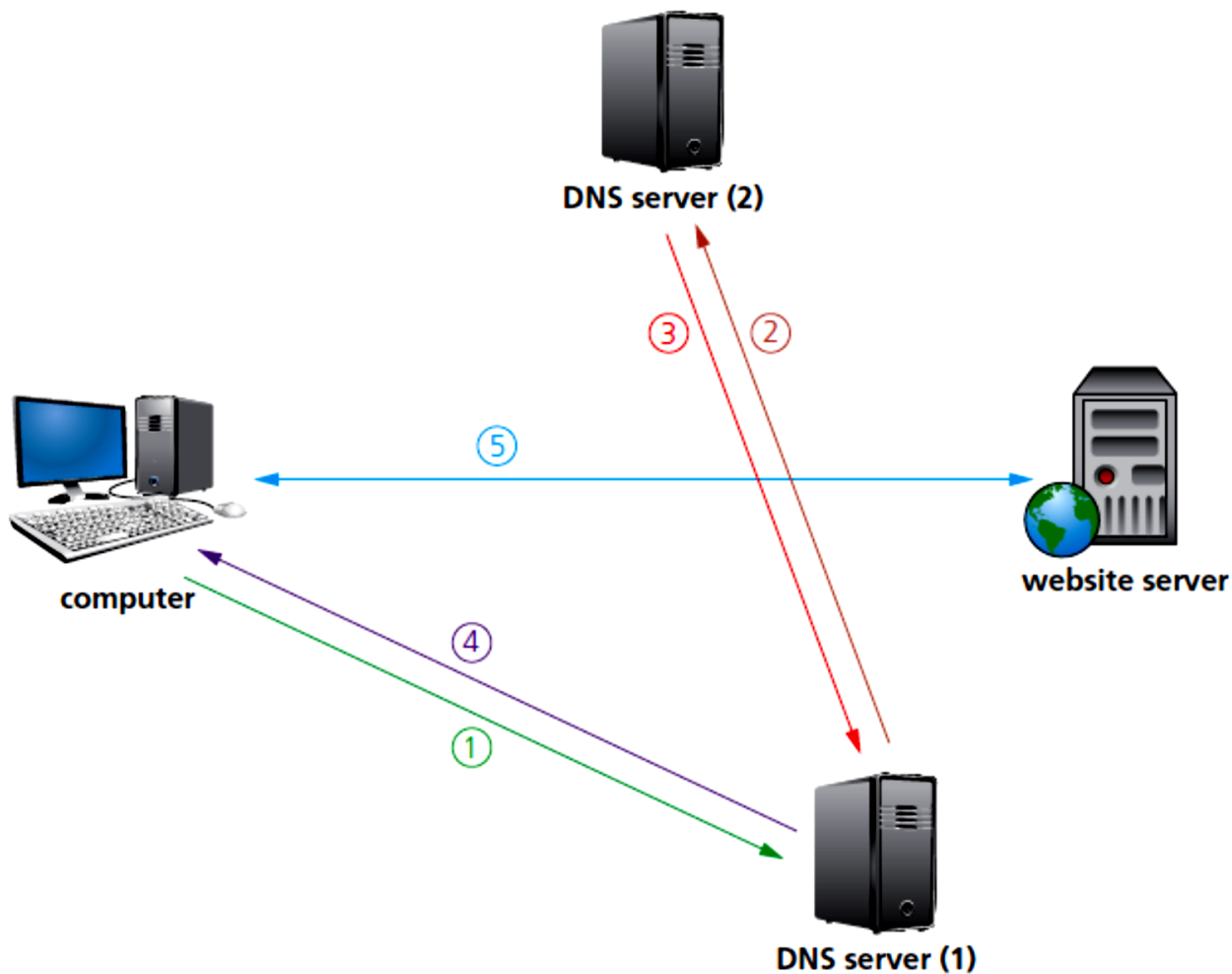
## Web browsers

- Browsers interpret (translate) the HTML from websites and show the result of the translation; for example, videos, images/text and audio.
- Most browsers have the following features:
  - » they have a home page.
  - » they can store a user's favourite websites/web pages (referred to as bookmarks).
  - » they keep a history of websites visited by the user (user history).
  - » they have the ability to allow the user to navigate forwards and backwards through websites/web pages already opened.
  - » many web pages can be open at the same time by using multiple tabs.
  - » they make use of cookies.
  - » they make use of hyperlinks that allow navigation between websites and web pages; links can be opened in one of two ways: *either* open in a new tab by using <ctrl> + <click> *or* open in the same tab by simply clicking on the link  
[www.hoddereducation.com](http://www.hoddereducation.com)
  - » data is stored as a cache (see Section 5.1.5)
  - » make use of JavaScript
  - » they use an address bar; for example:

## Retrieval and location of web pages

- HTML (HyperText Markup Language) is a language used to display content on browsers.
- All websites are written in HTML and hosted on a web server that has its own IP address.
- To retrieve pages from a website your browser needs to know this IP address.
- The **Domain Name Server (DNS)** (also known as domain name system) is a system for finding IP addresses for a domain name given in a URL.
- URLs and domain name servers eliminate the need for a user to memorise IP addresses.
- The DNS process involves converting a URL (such as [www.hoddereducation.co.uk](http://www.hoddereducation.co.uk)) into an IP address the computer can understand (such as 107.162.140.19).
- The DNS process involves more than one server.
- DNS servers contain a database of URLs with the matching IP addresses.

Figure 5.3 shows how a web page can be located and then sent back to the user's computer. The DNS plays a vital role in this process:



## How DNS is used to locate and retrieve a web page

- The user opens their browser and types in the URL ([www.hoddereducation.co.uk](http://www.hoddereducation.co.uk)) and the browser asks the DNS server (1) for the IP address of the website.
- In this case, let's assume the DNS server can't find [www.hoddereducation.co.uk](http://www.hoddereducation.co.uk) in its database or its cache, so it sends out a request to a DNS server (2).
- The DNS server (2) finds the URL and can map it to 107.162.140.19; this IP address is sent back to the DNS server (1) which now puts this IP address and associated URL into its cache/database.
- This IP address is then sent back to the user's computer.
- The computer now sets up a communication with the website server and the required pages are downloaded. HTML files are sent from the website server to the computer.
- The browser interprets the HTML, which is used to structure content, and then displays the information on the user's computer.
- (Note: in this case, the IP address was found on the second DNS server).

## Cookies

- Cookies are small files or code stored on a user's computer.
- They are sent by a web server to a browser on a user's computer.
- Each cookie is effectively a small look-up table containing pairs of (key, data) values, for example, (surname, Jones) (music, rock).
- Every time a user visits a website, it checks if it has set cookies on their browser before.
- If so, the browser reads the cookie which holds key information on the user's preferences such as language, currency and previous browsing activity.
- Cookies allow user tracking and maintain user preferences.
- Collected data can also be used to customise the web page for each individual user.
- For example, if a user buys a book online, the cookies remember the type of book chosen by the user and the web page will then show a message such as "Customers who bought Hodder IGCSE ICT also bought Hodder IGCSE Computer Science". **There are two types of cookie**
- Session cookie
- Persistent (or permanent) cookie.

## Session cookies

1. Session cookies are used, for example, when making online purchases.
2. They keep a user's items in a virtual shopping basket.
3. This type of cookie is stored in temporary memory on the computer, doesn't actually collect any information from the user's computer and doesn't personally identify a user.
4. Session cookies cease to exist on a user's computer once the browser is closed or the website session is terminated.
5. Session cookies are temporarily stored in RAM and not actually saved to your hard drive.
6. They are automatically deleted once you have finished your session browsing a website and you close the web browser.
7. Online shopping baskets will usually make use of session cookies.

## Persistent (permanent) cookies

- Persistent cookies remember a user's log in details (so that they can authenticate the user's browser).
- They are stored on the hard drive of a user's computer until the expiry date is reached or the user deletes it.
- These cookies remain in operation on the user's computer even after the browser is closed or the website session is terminated.
- Their advantage is that they remove the need to type in login details every time a certain website is visited. Some websites use cookies to store more personal information or user preferences.
- Many countries have introduced laws to protect users and these cookies are supposed to become deactivated after six months (even if the expiry date has not yet been reached).
- Persistent cookies are a very efficient way of carrying data from one website session to another, or even between sessions on related websites; they remove the need to store massive amounts of data on the web server itself.
- Storing the data on the web server without using cookies would also make it very difficult to retrieve a user's data without requiring the user to log in every time they visit the website.

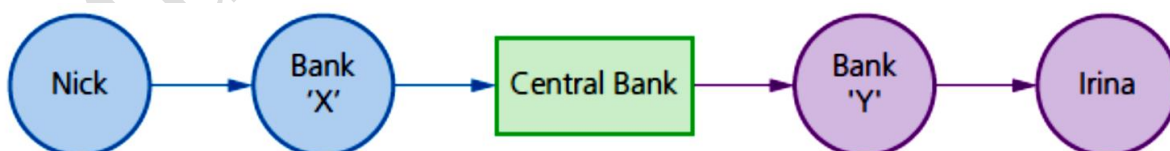
### Summary of the uses of (persistent) cookies:

- Allow the website to remember users' passwords, email addresses and invoice details, so they won't have to insert all of this information every time they visit or every time they purchase something from that website.
- Serve as a memory, enabling the website to recognise users every time they visit it.
- Save users' items in a virtual shopping basket/cart.
- Track internet habits and users' website histories or favourites/bookmarks.
- Target users with advertising that matches their previous buying or surfing habits.
- Store users' preferences (for example, recognise customised web pages).
- Are used in online financial transactions.
- Allow progress in online games and quizzes to be stored.
- Allow social networking sites to recognise certain preferences and browsing histories.
- Allow different languages to be used on the web pages automatically as soon as users log on.

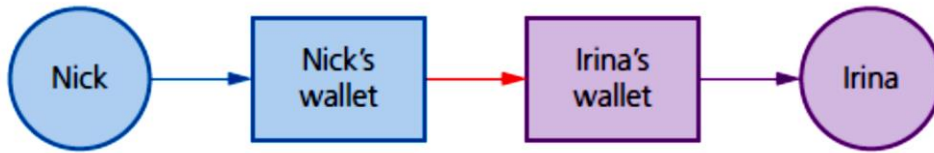
## Digital currency

### What is digital currency?

- **Digital currency** exists purely in a digital format.
- It has no physical form unlike conventional **fiat currency** (for example, \$, £, €, and ¥).
- Conventional currency is backed by governments and banks, it is referred to as fiat currency.)
- Digital currency is an accepted form of payment to pay for goods or services.
- As with cash or credit/debit cards, digital currency can be transferred between various accounts when carrying out transactions.
- It has made it possible to bank online (for example, using *PayPal*) or via a smartphone app (for example, *Apple Pay*).
- This is all possible because money only exists as data on a computer system, but it can be transferred into physical cash if we need it.
- Digital currency relies on a **central banking system**.
- For example, suppose Nick wishes to send Irina some money; Nick uses bank 'X' and Irina uses bank 'Y':



- The problem with centralisation is maintaining confidentiality and security; these have always been issues with digital currency systems.
- However, one example of digital currency, known as cryptocurrency, has essentially overcome these issues by introducing decentralisation:



## Cryptocurrency Currency

- Cryptocurrency uses **cryptography** to track transactions; it was created to address the problems associated with the centralisation of digital currency.
- Traditional digital currencies are regulated by central banks and governments (in much the same way as fiat currencies).
- This means all transactions and exchange rates are determined by these two bodies.
- Cryptocurrency has no state control and all the rules are set by the cryptocurrency community itself.
- Unlike existing digital currencies, cryptocurrency transactions are publicly available and therefore all transactions can be tracked and the amount of money in the system is monitored.
- The cryptocurrency system works by being within a **blockchain** network which means it is much more secure.

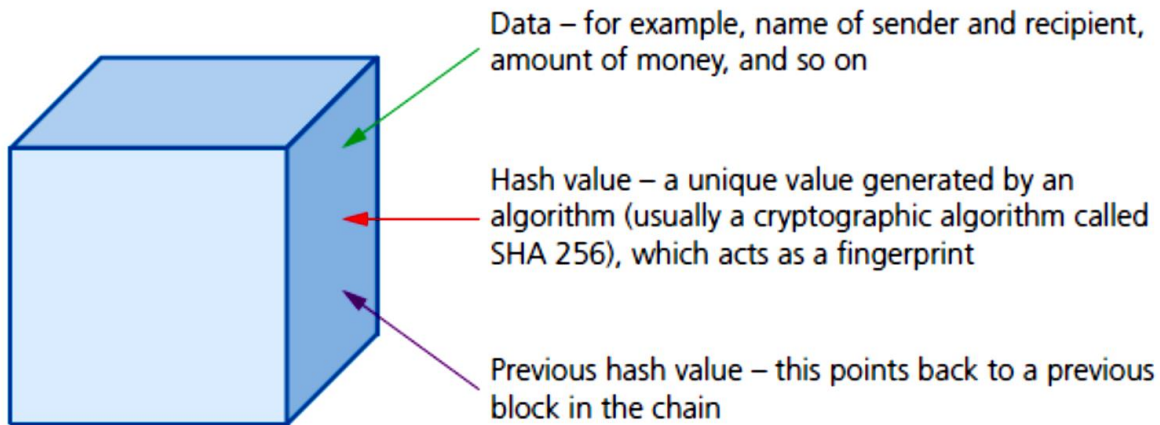
## Blockchaining

- Blockchain is a decentralised database.
- All the transactions of networked members are stored on this database.
- Essentially, the blockchain consists of a number of interconnected computers but they are **not connected** to a central server.
- All transaction data is stored on **all** computers in the blockchain network.
- Whenever a new transaction takes place, all the networked computers get a copy of the transaction; therefore **it cannot be changed without the consent** of **all** the network members.
- This effectively removes the risk of security issues such as hacking.
- Blockchain is used in many areas, such as:
  - » cryptocurrency (digital currency) exchanges
  - » smart contracts
  - » research (particularly within pharmaceutical companies)
  - » politics
  - » education.



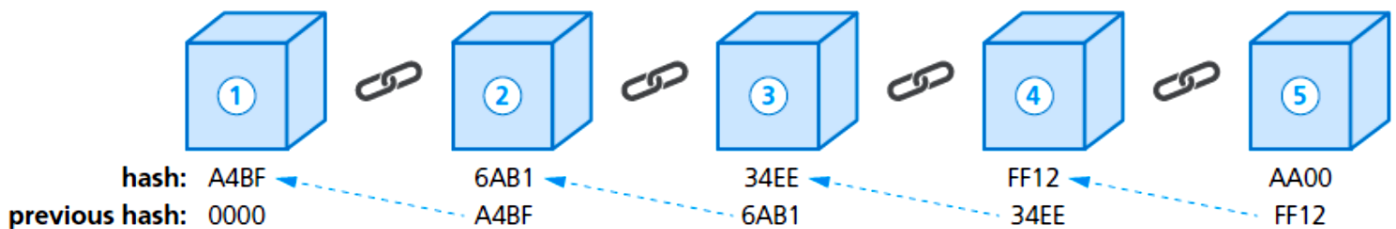
## How blockchain works

Whenever a new transaction takes place, a new **block** is created:



▲ **Figure 5.8** Block description

- A new hash value is created each time a new block is created.
- This hash value is unique to each block and includes a **timestamp**, which identifies when an event actually takes place.
- We will now consider what happens when a chain of blocks is created.
- Figure 5.9 shows part of a typical blockchain:



▲ **Figure 5.9** Part of blockchain (showing 5 blocks)

- It is clear from Figure 5.9 how these blocks are connected. Block '1' is known as the **genesis block** since it doesn't point to any previous block.
- Now suppose block '2' is changed in some way. Any changes to the data within block '2' will cause the value of the hash to change (it will no longer have the value 6AB1).
- This means that block '3' and beyond will now be invalid since the chain was broken between block '2' and '3' (previous hash 6AB1 in block '3' is no longer valid).
- This will prevent tampering (for example, by a hacker).
- However, it may be crossing your mind that computers are now so fast that it should be possible to quickly create a whole new string of blocks, and therefore recreate a new chain before the problem has been discovered.



- This is prevented by **proof-of-work**, which makes sure it takes ten minutes to determine the necessary proof-of-work for *each* block **before** it can be added to the chain.
- This is 'policed' by **miners**, which are special network users that get a commission for each new block created.
- Resulting whole process of creating new blocks is slowed down which foils hackers and also means that the currency is regulated by all the network computers.
- Consequently, this makes it almost impossible to hack into the blockchain since it would be necessary to attack every single block in the chain at the same time.
- It only takes one block to break the link for any transaction to be terminated.
- When a new block is created, it is sent to each computer in the blockchain and is checked for correctness before being added to the blockchain.
- If a new network user is created, they get a copy of everything in the whole blockchain system.

## Cyber security

### Cyber security threats

The following list shows the cyber threats which will be considered in this section:

#### Brute force attacks

If a hacker wants to 'crack' your password, they can systematically try all the different combinations of letters, numbers and other symbols until eventually they find your password. This is known as a **brute force attack**.

#### Data Interception

**Data interception** is a form of stealing data by tapping into a wired or wireless communication link. The intent is to compromise privacy or to obtain confidential information.

Interception can be carried out using a **packet sniffer**, which examines data packets being sent over a network.

Wi-Fi (wireless) data interception can be carried out using **wardriving** (or sometimes called **Access Point Mapping**). Using this method, data can be intercepted using a laptop or smartphone, antenna and a GPS device (together with some software) outside a building or somebody's house.

to safeguard against wardriving, the use of a **wired equivalency privacy (WEP)** encryption protocol, together with a firewall, is recommended. It is also a good idea to protect the use of the wireless router by having complex passwords

#### Distributed denial of service (DDoS) attacks

A **denial of service (DoS)** attack is an attempt at preventing users from accessing part of a network, notably an internet server.

The attacker may be able to prevent a user from:

- » accessing their emails
- » accessing websites/web pages
- » accessing online services (such as banking)

*One method of attack is to flood the network with useless **spam** traffic. How does this cause a problem?*

When a user enters a website's URL in their browser, a request is sent to the web server that contains the website or web page. Obviously, the server can only handle a finite number of requests. So if it becomes overloaded by an attacker sending out thousands of requests, it won't be able to service a user's legitimate request. This is effectively a denial of service.

### **In a distributed denial of service**

**(DDoS)** the spam traffic originates from many different computers, which makes it hard to block the attack.

An individual user or a website can guard against these attacks to some degree by:

- » using an up-to-date malware checker
- » setting up a firewall to restrict traffic to and from the web server or user's computer
- » applying email filters to filter out unwanted traffic (for example, spam).

There are certain signs a user can look out for to see if they have become a victim of a DDoS attack:

- » slow network performance (opening files or accessing certain websites)
- » inability to access certain websites
- » large amounts of spam email reaching the user's email account

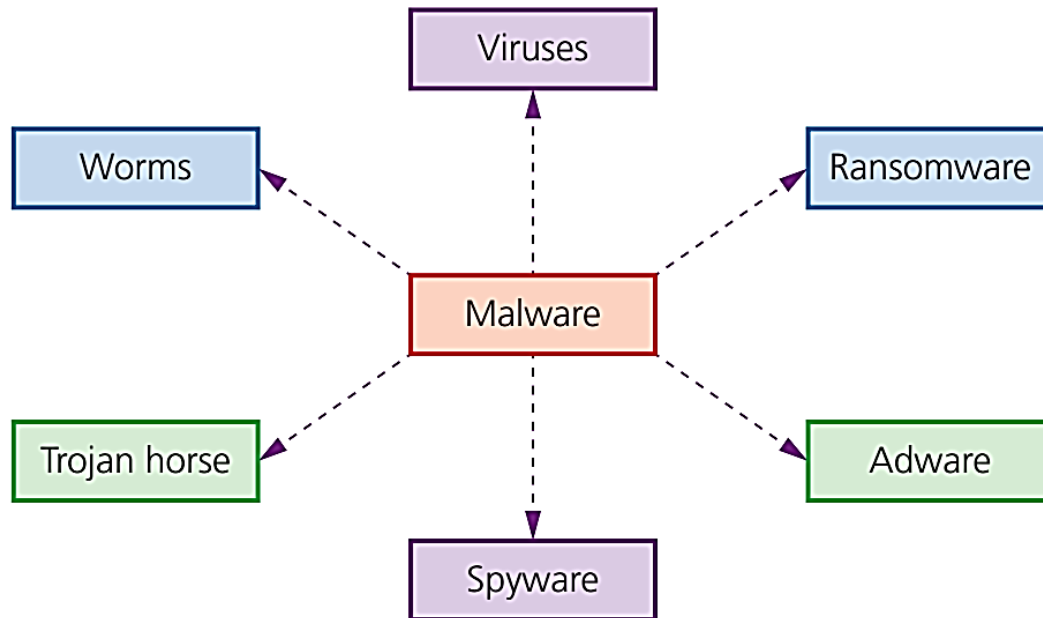
## **Hacking**

**Hacking** is generally the act of gaining illegal access to a computer system without the user's permission. This can lead to identity theft or the gaining of personal information; data can be deleted, passed on, changed or corrupted

Hacking can be prevented through the use of firewalls, user names and frequently changed strong passwords. Anti-hacking software and intrusion-detection software also exists in the fight against hacking.

**Ethical hacking.** This occurs when companies authorise paid hackers to check out their security measures and test how robust their computer systems are to hacking attacks.

**Malware** (viruses, worms, Trojan horse, spyware, adware and ransomware)



### Viruses

**Viruses** are programs or program code that replicate (copies themselves) with the intention of deleting or corrupting files, or causing a computer to malfunction

### Phishing

**Phishing** occurs when a cybercriminal sends out legitimate-looking emails to users.

There are numerous ways to help prevent phishing attacks:

- » users need to be aware of new phishing scams; those people in industry or commerce should undergo frequent security awareness training to become aware of how to identify phishing (and pharming) scams
- » it is important not to click on any emails links unless totally certain that it is safe to do so; fake emails can often be identified by 'Dear Customer .....' or 'Dear email person@gmail.com .....' and so on
- » it is important to run anti-phishing toolbars on browsers (this includes tablets and mobile phones) since these will alert the user to malicious websites contained in an email
- » always look out for https or the green padlock symbol in the address bar
- » regular checks of online accounts are also advisable as well as maintaining passwords on a regular basis
- » ensure an up-to-date browser is running on the computer device (which contains all of the latest security upgrades) and run a good firewall in the background at all times; a combination of a desktop firewall (usually software) and a network firewall (usually hardware) considerably reduces the risk of hacking, pharming and phishing on network computers
- » be very wary of pop-ups and use the browser to block them; if pop-ups get through your defences, don't click on 'cancel' since this can ultimately lead to phishing or pharming sites – the best option is to select the small in the top right-hand corner of the pop up window which closes it down.

**Spear phishing**; this is where the cybercriminal targets specific individuals or companies to gain access to sensitive financial information or industrial espionage – regular phishing is not specific regarding who the victims are.

## Pharming

**Pharming** is malicious code installed on a user's computer or on an infected website. The code redirects the user's browser to a fake website without the user's knowledge.

### DNS cache poisoning

Every time a user types in a URL, their browser contacts the DNS server; the IP address of the website will then be sent back to their browser. However, DNS cache poisoning changes the real IP address values to those of the fake website; consequently, the user's computer will connect to the fake website

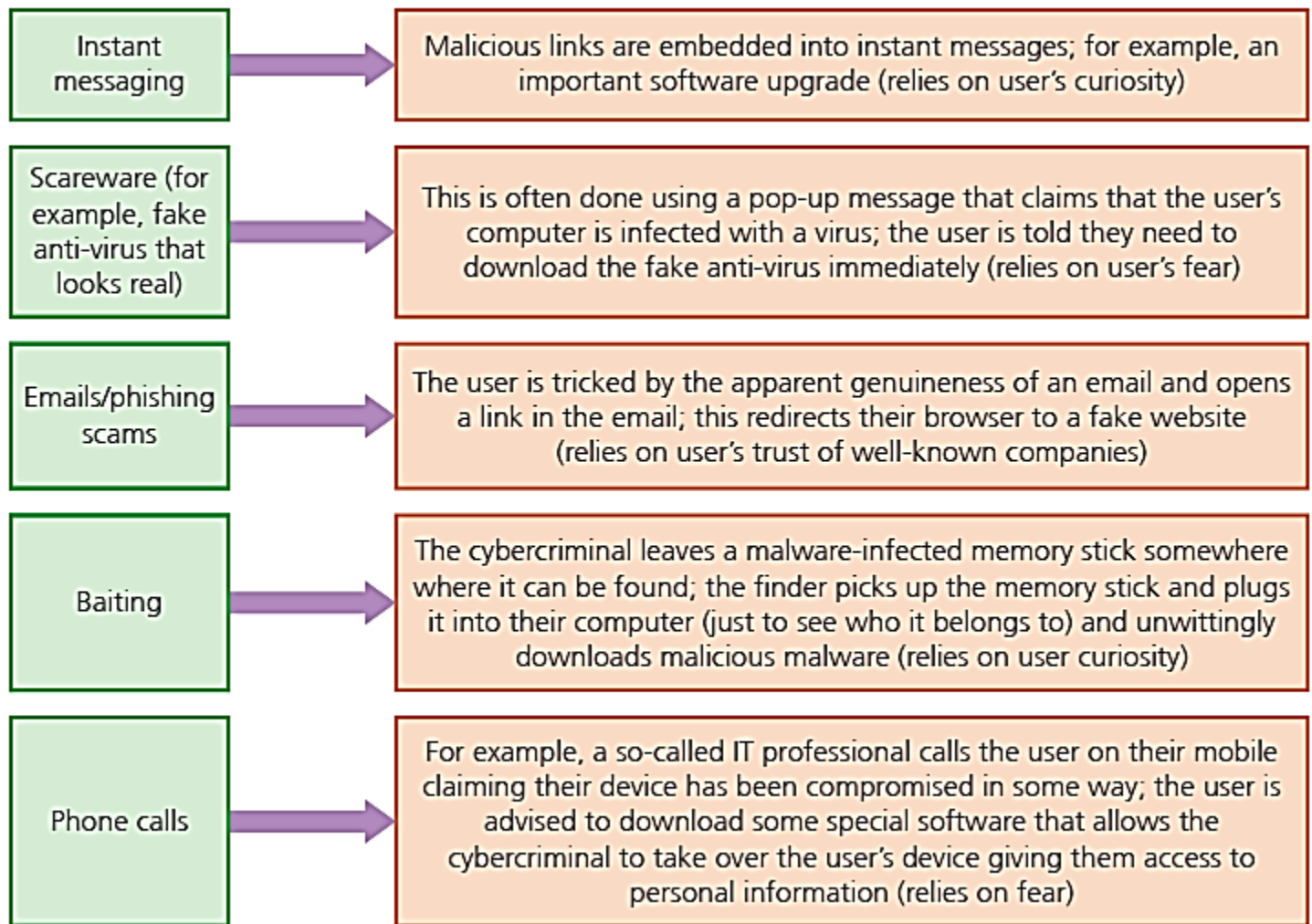
It is possible to mitigate against the risk of pharming:

- » Use of anti-virus software can detect unauthorised alterations to a website address and warn the user of the potential risks.
- » However, if the DNS server itself has been infected (rather than the user's computer) it is much more difficult to mitigate the risk.
- » Many modern browsers can alert users to pharming and phishing attacks.
- » It is very important to check the spelling of websites to ensure the web address used is correct.
- » As with phishing, use of **https** or the green padlock symbol in the address bar is an additional form of defence.

### » social engineering

**Social engineering** occurs when a cybercriminal creates a social situation that can lead to a potential victim dropping their guard.

There are five types of threat that commonly exist:.



The whole idea is based on the exploitation of certain human emotions; the three most common ones to exploit are:

- » fear – the user is panicked into believing their computer is in immediate danger and isn't given time to logically decide if the danger is genuine or not; fear is a very powerful emotion that can easily be exploited by a cybercriminal
- » curiosity – the user can be tricked into believing they have won a car or they find an infected memory stick lying around; their curiosity gets the better of them and they give their details willingly to win the car (for example, credit card details to pay for delivery or road tax) or they are curious who the memory stick belongs to; without thinking clearly, their curiosity gets the better of them and the damage is done.
- » empathy and trust – a real belief that all genuine-sounding companies can be trusted, therefore emails or phone calls coming from such companies must be safe; a dangerous assumption that the cybercriminal can exploit fully.

## Keeping data safe from security threats

### Access levels

For example, in a hospital it would not be appropriate for a cleaner to have access to medical data about a patient. However, a consultant would need access to this vital data. Therefore, most systems have a hierarchy of access levels depending on a person's level of security; this is usually achieved using a user name and password.

When using databases, levels of access are particularly important; it is essential to determine who has the right to read, write and delete data, for example. By having different views of data tables, it is possible for different users to only have access to certain data.

There are usually four access levels in social media like Facebook:

- Public access (this refers to the data anyone from the general public can access)
- Friends (only people identified as 'friends' by the owner of the data can see certain data)
- Custom (this allows the user to further refine what data can be seen by 'friends' allowing them to exclude certain content from selected people)
- Data owner (this is data only the owner of the data can see).

In this type of application, users are allowed to use **privacy settings** rather than passwords to decide the level of access.

### Anti-malware

The two most common types of anti-malware are anti-virus and anti-spyware.

### Anti-spyware

**Anti-spyware** software detects and removes spyware programs installed illegally on a user's computer system.

The software is based on one of the following methods:

- » rules – in this case, the software looks for typical features which are usually associated with spyware thus identifying any potential security issues.
- » file structures – in this case, there are certain file structures associated with potential spyware which allows them to be identified by the software.

Anti-spyware is now often part of a generic malware bundle that contains an anti-virus, anti-spyware and a personal firewall.

**The general features of anti-spyware are:**

- » detect and remove spyware already installed on a device
- » prevent a user from downloading spyware
- » encrypt files to make the data more secure in case it is 'spied' on
- » encryption of keyboard strokes to help remove the risk posed by the keylogging aspects of some spyware
- » blocks access to a user's webcam and microphone (the software stops the spyware taking over the control of a user's webcam and microphone which can be used to collect information without the user's knowledge)
- » scans for signs that the user's personal information has been stolen and warns the user if this has happened.

**Authentication**

**Authentication** refers to the ability of a user to prove who they are. There are three common factors used in authentication:

- » something you know (for example, a password or PIN code)
- » something you have (for example, a mobile phone or tablet)
- » something which is unique to you (for example, biometrics).

There are a number of ways authentication can be done.

**Passwords and user names**

Passwords are used to restrict access to data or systems.

They should be hard to crack and changed frequently to retain any real level of security.

Passwords can also take the form of biometrics (for example, on a mobile phone – see later).

In addition to protecting access levels to computer systems, passwords are frequently used when accessing the internet. For example:

- » when accessing email accounts
- » when carrying out online banking or shopping
- » accessing social networking sites.



It is important that passwords are protected; some ways of doing this are described below:

- » run anti-spyware software to make sure that your passwords aren't being relayed back to whoever put the spyware on your computer
- » change passwords on a regular basis in case they have come into the possession of another user, illegally or accidentally
- » passwords should not be easy to crack (for example, your favourite colour, name of a pet or favourite music artist); passwords are grouped as either strong (hard to crack or guess) or weak (relatively easy to crack or guess)
- » strong passwords should contain:
  - at least one capital letter
  - at least one numerical value
  - at least one other keyboard character (such as @, \*, &, etc.)
  - an example of a strong password would be: Sy12@#TT90kj=0
  - an example of a weak password would be: GREEN

When the password is typed in, it often shows on the screen as \*\*\*\*\* so nobody else can see what the user has typed in. If the user's password doesn't match up with the user name, then access will be denied.

Many systems ask for a new password to be typed in twice as a verification check (to check for input errors). To help protect the system, users are only allowed to type in their password a finite number of times – usually three times is the maximum number of tries allowed before the system locks the user out. After that, the user will be unable to log on until they have reset their password.

When using an online company, if a user forgets their password or they need to reset it, they will be sent an email which contains a link to a web page where they can reset their password. This is done as an added precaution in case an unauthorised person has tried to change the user's password.

As mentioned above, it is usually necessary to use a user name as well as a password. This gives an additional security level since the user name and password must match up to allow a user to gain access to, for example, a bank website.

### Biometrics

**Biometrics** can be used in much the same way as passwords as a way of identifying a user. Biometrics relies on certain unique characteristics of human beings; examples include:

- » fingerprint scans
- » retina scans
- » face recognition
- » voice recognition.

Biometrics is used in a number of applications as a security device.

For example, some of the latest mobile phones use fingerprint matching before they can be operated; some pharmaceutical companies use face recognition or retina scans to allow entry to secure areas.

We will now consider fingerprint scanning and retina scans in a little more detail.

### ***Fingerprint scans***

Images of fingerprints are compared against previously scanned fingerprint images stored in a database; if they match, then a user has been correctly recognised.

The system compares patterns of 'ridges' and 'valleys' that are unique.

The accuracy of the scan is about around 1 in 5000.

Fingerprint scanning techniques have the following benefits as a form of security:

- » fingerprints are unique, therefore this technique can improve security since it would be difficult to replicate a person's fingerprints
- » other security devices (such as magnetic cards to gain entry to a building) can be lost or even stolen which makes them less effective.
- » it would be impossible to 'sign in' for somebody else since the fingerprints would match with only one person on the database.
- » fingerprints can't be misplaced; a person always has them!

What are the drawbacks of fingerprint scanning?

- » it is relatively expensive to install and set up
- » if a person's fingers are damaged through an injury, this can have an effect on the scanning accuracy
- » some people may regard any biometric device as an infringement of civil liberties.

### ***Retina scans***

Retina scans use infrared light to scan the unique pattern of blood vessels in the retina (at the back of the eye);

It is a rather unpleasant technique requiring a person to sit totally still for 10 to 15 seconds while the scan takes place.

It is very secure since nobody has yet found a way to duplicate the blood vessels patterns. The accuracy is about 1 in 10 million.

▼ **Table 5.3** Comparison of biometric devices

| Biometric technique | Benefits                                                                                                                                         | Drawbacks                                                                                                                                                                                                 |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fingerprint scans   | it is one of the most developed biometric techniques<br>very easy to use<br>relatively small storage requirements for the biometric data created | for some people it is very intrusive, since it is still related to criminal identification<br>it can make mistakes if the skin is dirty or damaged (e.g. cuts)                                            |
| retina scans        | very high accuracy<br>there is no known way to replicate a person's retina                                                                       | it is very intrusive<br>it can be relatively slow to verify retina scan with stored scans<br>very expensive to install and set up                                                                         |
| face recognition    | non-intrusive method<br>relatively inexpensive technology                                                                                        | it can be affected by changes in lighting, the person's hair, change in age, and if the person is wearing glasses                                                                                         |
| voice recognition   | non-intrusive method<br>verification takes less than 5 seconds<br>relatively inexpensive technology                                              | a person's voice can be easily recorded and used for unauthorised access<br>low accuracy<br>an illness such as a cold can change a person's voice, making absolute identification difficult or impossible |

### Two-step verification

**Two-step verification** requires two methods of authentication to verify who a user is.

It is used predominantly when a user makes an online purchase using a credit/debit card as payment method.

For example, suppose Kate wishes to buy a new camera from a website. She logs into the website using her computer. This requires her to enter a user name and a password, which is step 1 of the authentication process.

To improve security, an eight-digit PIN (called a one-time pass code) is sent back to her either in an email or as a text message to her mobile phone (the mobile phone has already been registered by Kate on the website as the second stage of the authentication process).

Kate now enters this eight-digit PIN into her computer and she is now authorised to buy the camera.

### Automatic software updates

Automatic software updates mean software on computers and mobile phones/tablets is kept up-to-date. Sometimes this is done overnight or when you log off the device.

These updates are vital since they may contain **patches** that update the software security (to protect against malware) or improve the software performance (for example, removal of bugs and addition of new features).

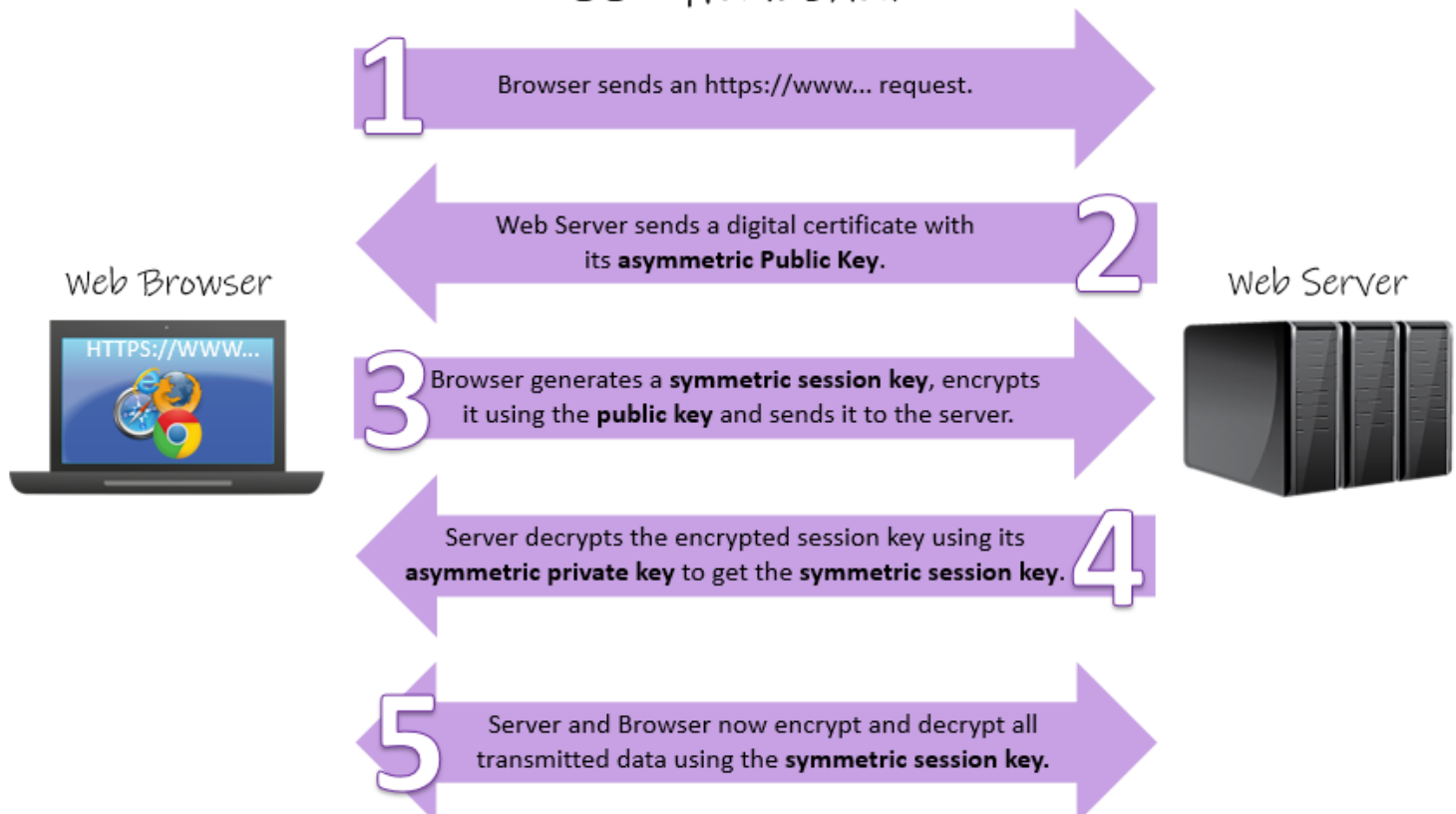
The only downside to this is the potential for updates to disrupt your device following installation. If this happens, the user either has to wait for another patch to put this right, or use the techniques described in Chapter 4 that reverse the clock time to an earlier date before the updates were made.

Checking the spelling and tone of communication and URL links

When emails are sent to you, there are three actions you always need to take before opening them or activating any links in them.

- » Check out the spellings in the email and in the links; professional, genuine organisations will not send out emails which contain spelling or major grammatical errors (for example, Amazzon.com)
- » Carefully check the tone used in the email message; if it is rushing you into doing something or if the language used seems inappropriate or incorrect, then it could be a phishing email or worse.

## SSL Handshake



To recap, the 5 steps of a SSL handshake are:

1. Browser sends an `https://www...` request.
2. Web Server sends a digital certificate with its asymmetric Public Key.
3. Browser generates a symmetric session key, encrypts it using the public key and sends it to the server.
4. Server decrypts the encrypted session key using its asymmetric private key to get the symmetric session key.
5. Server and Browser now encrypt and decrypt all transmitted data using the symmetric session key. The communication channel is secure as only the web browser and the server know the symmetric session key. The session key is only used for that session. If the browser was to connect to the same server at another time, a new session key would be created following the 5 steps of the SSL handshake.

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